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Clinical Study

GPT4LFS (generative pretrained transformer 4 omni for lumbar foramina stenosis): enhancing lumbar foraminal stenosis image classification through large multimodal models

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Abstract

BACKGROUND CONTEXT: Lumbar foraminal stenosis (LFS) is a common spinal condition that requires accurate assessment. Current magnetic resonance imaging (MRI) reporting processes are often inefficient, and while deep learning has potential for improvement, challenges in generalization and interpretability limit its diagnostic effectiveness compared to physician expertise.

PURPOSE: The present study aimed to leverage a multimodal large language model to improve the accuracy and efficiency of LFS image classification, thereby enabling rapid and precise automated diagnosis, reducing the dependence on manually annotated data, and enhancing diagnostic efficiency.

STUDY DESIGN/SETTING: Retrospective study conducted from April 2017 to March 2023.

PATIENT SAMPLE: Sagittal T1-weighted MRI data for the lumbar spine were collected from 1,200 patients across 3 medical centers. A total of 810 patient cases were included in the final analysis, with data collected from 7 different MRI devices.

OUTCOME MEASURES: Automated classification of LFS using the multi modal large language model. Accuracy, sensitivity, Specificity and Cohen's Kappa coefficient were calculated.

METHODS: An advanced multimodal fusion framework GPT4LFS was developed with the primary objective of integrating imaging data and natural language descriptions to comprehensively capture the complex LFS features. The model employed a pretrained ConvNeXt as the image processing module for extracting high-dimensional imaging features. Concurrently, medical descriptive texts generated by the multimodal large language model GPT-4o and encoded and feature-extracted using RoBERTa were utilized to optimize the model's contextual understanding capabilities. The Mamba architecture was implemented during the feature fusion stage, effectively integrating imaging and textual features and thereby enhancing the performance of the classification task. Finally, the stability of the model's detection results was validated by evaluating classification task metrics, such as the accuracy, sensitivity, specificity, and Kappa coefficients.

FDA device/drug status: Not applicable.

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RESULTS: The training set comprised 6,299 images from 635 patients, the internal test set included 820 images from 82 patients, and the external test set was composed of 930 images from 93 patients. The GPT4LFS model demonstrated an overall accuracy of 93.7%, sensitivity of 95.8%, and specificity of 94.5% in the internal test set (Kappa=0.89, 95% confidence interval (CI): 0.84−0.96, p<.001). In the external test set, the overall accuracy was 92.2%, with a sensitivity of 92.2% and a specificity of 97.4% (Kappa=0.88, 95% CI: 0.84−0.89, p<.001). Both the internal and external test sets showed excellent consistency in the model. The code is freely accessible on GitHub at the following repository: https://github.com/ElzatElham/GPT4LFS.

CONCLUSIONS: Using the GPT4LFS model for LFS image categorization demonstrated accuracy and the capacity for feature description at a level commensurate with that of professional clinicians.

Keywords: Deep learning; Lumbar degeneration; Lumbar foraminal stenosis; Multimodal Large Language Model; Image Classification; Automated Diagnosis

Introduction

Lumbar foraminal stenosis (LFS) is a prevalent degenerative spinal condition, accounting for 8% to 11% of all lumbar spine disorders [1]. With the advancement of diagnostic techniques, the diagnosis rate of LFS has been observed to increase continuously [2]. An accurate assessment of stenosis degree is of paramount importance for the development of appropriate treatment plans, particularly for patients who require active intervention [1,3]. It is therefore incumbent upon radiologists to produce clinical reports that are both concise and accurate in order to convey the key findings from lumbar magnetic resonance imaging (MRI) in an efficacious manner [4]. However, the existing workflow is often unwieldy and repetitive and there is a pressing need for improvement.

In recent years, deep learning technology has gradually become a reliable support tool in orthopedic MRI diagnostics with the aim to address the efficiency issues in medical image analysis [5]. Won et al [6] employed a deep convolutional neural network for the objective grading of lumbar spinal stenosis. Their results revealed a high degree of consistency with the diagnoses of 2 expert reviewers (77.9% and 83.0%). Natalia et al [7] employed a combination of the support vector machine algorithm and the pretrained DenseNet201 model to determine the optimal MR image slice for detecting lumbar spinal stenosis. This approach demonstrated high accuracy (88.0%) and recall (88.0%). The study by Lewandrowski et al [8] showed that the deep learning neural network (RadBot) exhibited high sensitivity (72.0%) and specificity (83.0%) when detecting nerve element compression in lumbar MRI as compared to radiologist diagnoses. This indicates that deep learning technology is enhancing the precision and expediency of orthopedic diagnosis. Hallinan et al [2] developed a model for the 4-class classification task of lumbar MR images related to the spinal canal, lateral recess, and neural foraminal stenosis. The classification of spinal canal stenosis (Kappa coefficient=0.82) showed a high level of agreement with the radiologists' conclusions, with a slightly lower agreement in lateral recess stenosis (Kappa=0.72) and neural foraminal stenosis (Kappa=0.75). The study by Lim et al [9] demonstrated that deep learning-assisted interpretation significantly reduced the MRI diagnostic time for radiologists from 124 to 274 s to 47 to 71 s. The model displayed higher or comparable consistency in various stenosis gradings, particularly in the 4-class neural foraminal stenosis, with the Kappa value increasing from 0.39 to 0.71. These methods have played an important role in orthopedic auxiliary diagnostics. However, the reliance on single visual features has resulted in inadequate generalization ability and interpretability of the models, leading to a gap in diagnostic capability compared to that of physicians.

To overcome the singularity of visual models, researchers have begun to combine multimodal large language model (MLLM) to complete medical diagnostic tasks. Jiang et al [10] proposed the Text-Visual-Prompt SAM zero-shot algorithm combined with large language model generative pretrained transformer 4 (GPT-4), which achieved zero-shot segmentation without human intervention with an average Dice score exceeding 80% on nonradiological images. Yan et al [11,12] proposed that generative pretrained transformer 4 with vision (GPT-4V) can distinguish various types of queries in medical visual questions with an accuracy of 35.3%, which, although not reaching clinical standards, lays the foundation for artificial intelligence (AI)-assisted diagnosis and predicts further breakthroughs. Schubert et al demonstrated the diagnostic capability of GPT-4V in complex clinical scenarios, achieving a diagnostic accuracy of 80.6% in 93 cases from image challenge datasets [13,14]. Han et al used GPT-4V to achieve accuracies of 73.3% and 88.7% in 140 and 348 clinical cases, respectively, significantly outperforming human outcomes [13,15,16]. Furthermore, while GPT-4V has shown great potential in generating descriptive chest X-ray reports, its precision in identifying specific medical organs and signs still needs to be improved [17,18]. Overall, the widespread application of MLLM in the medical field represents significant potential for improving healthcare quality and efficiency.

In the present study, an MLLM generative pretrained transformer 4 omni for lumbar foramina stenosis (GPT4LFS) was specifically designed for LFS diagnosis. By incorporating prompts generated by generative pretrained transformer 4 omni (GPT-4o) [19], which has stronger performance than GPT-4V, and fusing multimodal features, the model was able to extract extensive knowledge representations. This approach reduced reliance on manual annotation and enhanced the comprehensiveness and accuracy of LFS diagnosis. Multicenter dataset testing demonstrated the model's excellent performance and generalization ability, showing its significant potential in assisting doctors with image reading and report writing.

Materials and methods

Patient population

The present study utilized retrospective analysis that involved 3 provinces. The training and internal test sets included 1,000 randomly selected patients from the Hospital A, while the external test set consisted of 200 randomly selected patients from the Hospital B and Hospital C. The inclusion criteria were patients aged 18 years and older who underwent lumbar MRI due to low back pain, with T1-weighted sagittal MR image data of the lumbar spine. Exclusion criteria comprised patients who previously underwent lumbar surgery, those with diseases unrelated to orthopedics, and data that did not meet MRI quality standards. The data were randomly extracted between April 2017 and March 2023 and were anonymized. Due to the retrospective nature of the study and data de-identification, all 3 centers were exempt from the requirement for informed consent.

Image processing

Multiple strategies were employed to minimize the impact of image variability to ensure the stability and consistency of the model across different devices. Based on our self-developed GPTSP system, we have achieved a fully automated processing workflow that transforms DICOM data into foraminal slices. This system automatically locates the foramina at each level from L_1 to S_1 using 3-dimensional DICOM data and extracts sagittal 2-dimensional slices for subsequent analysis. Before inputting the images into the GPT4LFS model for training and prediction, all images in the dataset were resized to 1,024 × 1,024 pixels and had their grayscale values and contrast adjusted (Fig. 1). Data augmentation techniques, such as flipping, rotation, and color jitter, were applied during training to increase the diversity of the data [20]. These strategies help to reduce the effects of inter-device image variability on data consistency, ensuring stable training outcomes.

Image annotation and result verification

The participating physicians classified the degree of LFS based on comparison images according to the recognized standards [9,21] into the following 4 levels: normal, mild, moderate, and severe. Three orthopedic surgeons (EE with 10 years of clinical experience, JS and HZ each with 20 years of clinical experience) classified the uniformly sized images to ensure annotation accuracy. The annotations were performed using the spinal imaging intelligent aided reading system (SY-GPTSP, SDSY/China). Orthopedic expert SF with 30 years of clinical experience evaluated the annotations in cases of discrepancy to ensure data accuracy.

Generating image descriptions using MLLM

Descriptions of MR lumbar foramen images were generated using MLLM (GPT-4o). These models combined image and text information to produce insightful outputs. The approach designed 2 different prompting strategies for each MR image to guide the model in generating accurate descriptions.

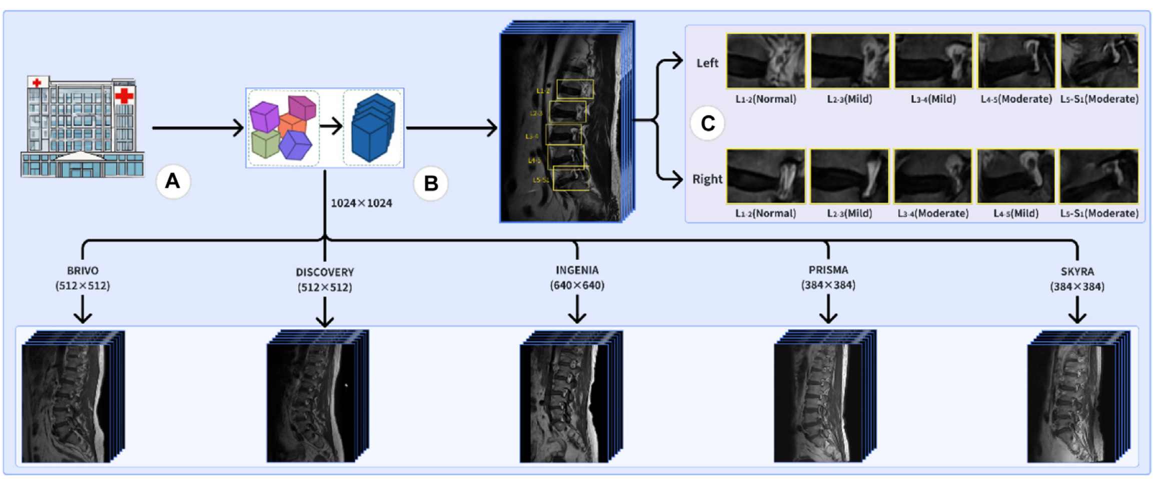

Fig. 1. Data processing workflow. (A) Data collection, (B) Data normalization, (C) Segmentation of the left and right nerve root regions in the sagittal MR images for different spinal levels.

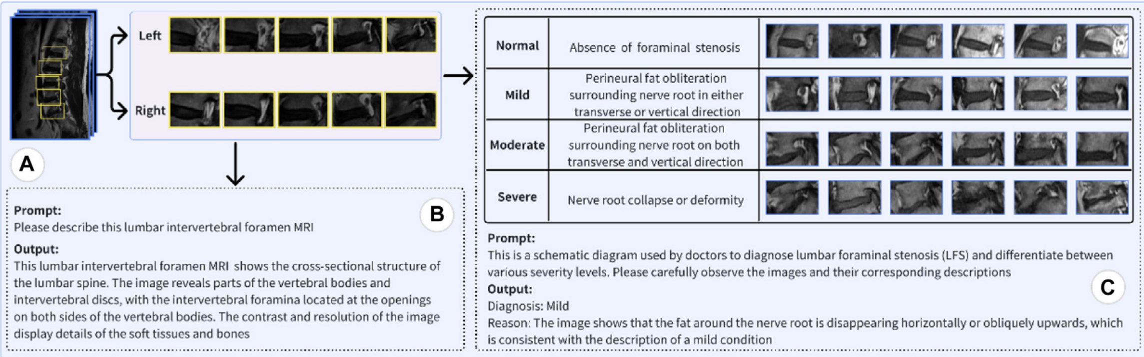

Fig. 2. Description generation example. (A) Different nerve root regions in the spinal segments were used as GPT-4o input. (B) Description generated by GPT-4o using the basic prompting method. (C) Description generated by GPT-4o using the optimized prompting strategy.

Image description generation strategies

Two differentiated prompting strategies were designed for each lumbar intervertebral foraminal MR image under analysis to guide the MLLM in generating more precise descriptions. The first strategy was a basic prompting method (Fig. 2B), which is direct and straightforward: “Please describe this lumbar intervertebral foramen MRI.” The second optimized prompting strategy (Fig. 2C) involving 2 critical steps was developed to enhance the precision and relevance of the descriptions. a). First, a standard schematic of LFS grading criteria was input into the MLLM, accompanied by a detailed contextual explanation as follows: “This is a schematic diagram used by doctors to diagnose LFS and differentiate between various severity levels. Please carefully observe the images and their corresponding descriptions.” b). A more targeted prompt was employed for the specific image to be classified: “Based on the previously shown LFS grading criteria, please provide a detailed description of this lumbar intervertebral foramina MR image, clearly stating the degree of LFS and the basis for your judgment.”

Description set construction and comparative evaluation

Two sets of different descriptions were generated for each training image based on the 2 designed prompting strategies. The performance of the different strategies was assessed by comparing these 2 sets of descriptions and their effects on subsequent classification tasks. Fig. 2 represents a typical example of the description generation, clearly showcasing the model’s output under specific prompts and intuitively revealing the effectiveness of the various strategies.

GPT4LES model

The present study adopted an advanced multimodal fusion framework that integrated imaging data and natural language descriptions to comprehensively capture the complex LFS features. Accurate LFS severity classification was achieved using unified preprocessing of the internal and external test sets.

The pretrained ConvNeXt [22] was used as the image processing module (Fig. 3C) to extract high-dimensional imaging features from the MR images. ConvNeXt was chosen for its excellent feature extraction capability and adaptability to medical images. Additionally, the model integrated the image descriptions generated by the MLLM (GPT-4o) (Fig. 3D). These descriptions were encoded and feature-extracted using the pretrained RoBERTa [23] language model (Fig. 3E), leveraging its strong contextual understanding and rich semantic representation to transform unstructured medical descriptions into high-dimensional semantic vectors. This allowed the implicit incorporation of the medical knowledge from the large language model into the classification process.

The Mamba architecture [24] was employed in the feature fusion stage, which is a sequence processing technique based on a state-space model. The state-space model is a mathematical framework used for handling time series data, capable of capturing the dynamic characteristics of the data. The Mamba architecture introduces a selective state space mechanism, providing more efficient computation and improved performance when processing sequential data. Compared to traditional Transformer models, Mamba exhibits significant advantages in handling long sequential data, better capturing long-term dependencies while reducing computational complexity. This characteristic allows it to effectively coordinate the spatiotemporal relevance between imaging features and textual semantic features during multimodal feature fusion. Mamba effectively integrated the imaging features and text features, enhancing the recognition and classification of the affected regions and providing a deeper understanding of the multidimensional LFS manifestations. This approach did not only improve the model’s ability to capture complex pathological features but also significantly enhanced the accuracy and reliability of the classification.

Finally, the imaging features extracted by ConvNeXt and the text features processed by RoBERTa were input into the Mamba architecture for fusion (Fig. 3F). The fused features were then fed into a fully connected layer for

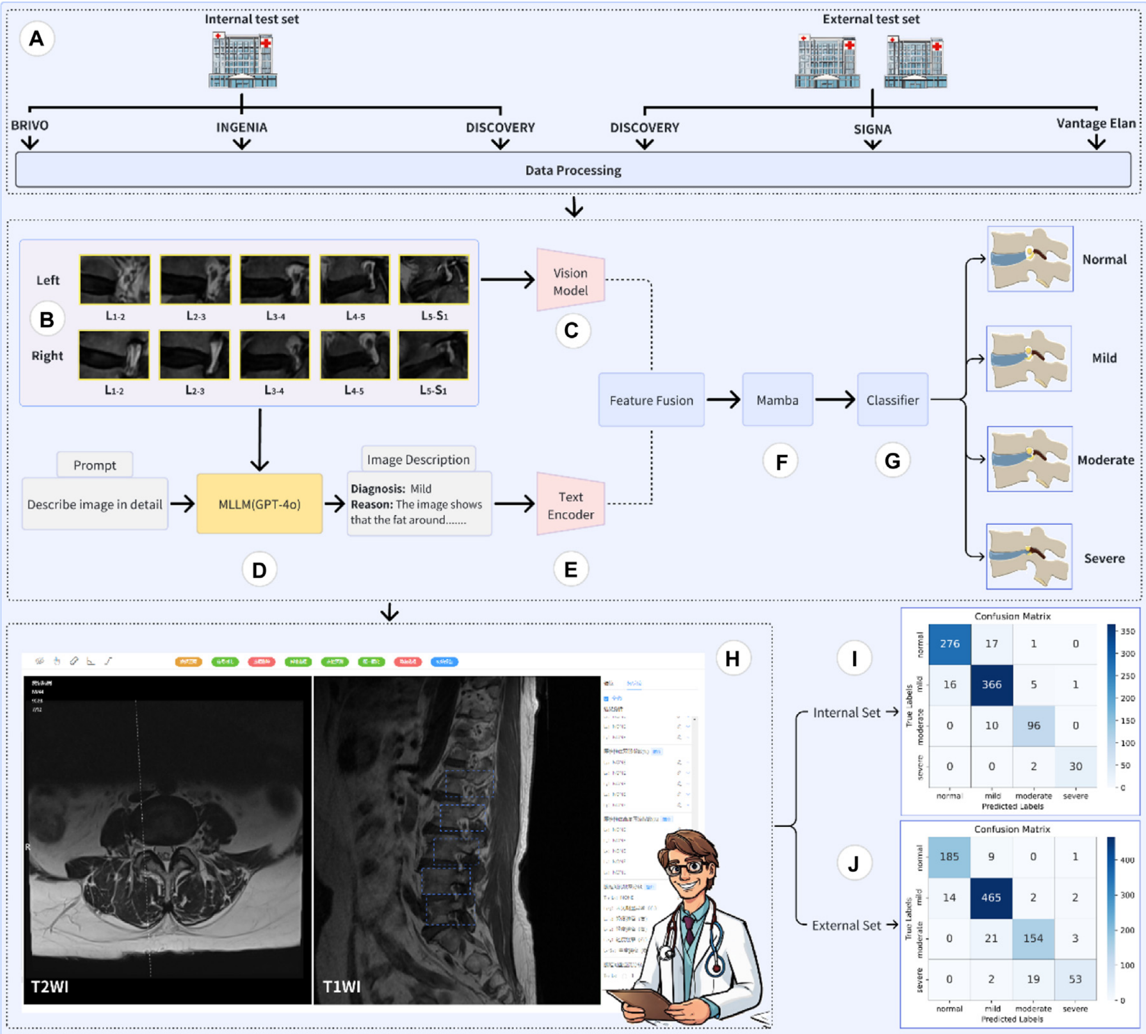

Fig. 3. GPT4LFS framework. (A) Both the internal and external test datasets underwent unified data normalization preprocessing. (B) Images of nerve root regions from different spinal segments. (C) Feature extraction from the original images using the ConvNeXt model. (D) Generation of image descriptions using the MLLM (GPT-4o). (E) Encoding and feature extraction of the descriptions generated by the large language model using RoBERTa. (F) Integration of the imaging and text features using the feature fusion in the Mamba module. (G) Classification into 4 stenosis categories using the fully connected layer. (H) Comparison of the model's classifications with the ground truth labels based on the physician's standards. (I) Visualization of the confusion matrices for the internal test set results. (J) Visualization of the confusion matrices for the external test set results.

classification (Fig. 3G). The cross-entropy loss function was used to calculate the difference between the predicted values and the true labels. The entire workflow was designed to be compact and efficient, ensuring both processing speed and accuracy and reliability of the results.

Statistical analysis

Continuous variables were presented as the mean±standard deviation (SD). The consistency between the model's prediction and manual annotation was evaluated using Cohen's Kappa coefficient (Kappa). Its consistency level was defined as follows: when Kappa=0, the consistency was entirely due to chance; when Kappa=1, the 2 classification results were completely consistent; when Kappa ≤0.4, the consistency was poor; when 0.4 <Kappa ≤0.6, the consistency was moderate; when 0.6<Kappa ≤0.8, the consistency was high; and when Kappa >0.8, the consistency was excellent. The significance level was p<.01. Statistical analysis was performed using Python (3.7.16/Python Software Foundation/USA).

Results

Patient characteristics

The final dataset included 8,049 T1-weighted images from 810 patients acquired using 7 different MRI devices. The training set consisted of 635 cases [mean age, 63.20±15.34 years [SD] [range, 18−99 years]), including 363 females (mean age, 62.92±14.69 years [SD] [range, 20−95 years]) and 269 males (mean age, 63.57 years±16.19 years [SD] [range, 18−99 years]). The internal test set included 82 cases, comprising 34 females (mean age, 46.50±

Table 1
Basic baseline distribution of patient datasets

Images	Patients	Age (mean±SD)	Sex (Male/Female)	Manufacturer	Device type
Training					
6,299	635	63.20±15.34	272/363	GE	Brivo MR355
				GE	Discovery MR750
				PHILIPS	Ingenia Elition X
				SIEMENS	Prisma
				SIEMENS	Skyra
Internal					
300	30	46.23±11.55	17/13	GE	Brivo MR355
240	24	51.92±9.85	14/10	GE	Discovery MR750
280	28	44.54±10.50	17/11	PHILIPS	Ingenia Elition X
External					
290	29	57.55±11.02	17/12	GE	Discovery MR750
320	32	57.53±12.77	18/14	GE	Signa Pioneer
320	32	55.03±14.51	16/16	TOSHIBA	Vantage Elan

SD, standard deviation.

12.93 years [SD] [range, 24−65 years]) and 48 males (mean age, 47.90±9.54 years [SD] [range, 29−64 years]). The external test set consisted of 93 cases, including 42 females (mean age, 56.71±13.34 years [SD] [range, 23−86 years]) and 51 males (mean age, 56.65±12.51 years [SD] [range, 32−76 years]) (Table 1).

Performance of different models under various description sets

In the present study, GPT-4o demonstrated exceptional performance in traditional benchmarking tasks related to text, reasoning, and programming abilities and achieved new high watermarks in multilingual capabilities and visual understanding [19]. The core innovation of the experiment was in leveraging the large language model to generate 2 sets of medical descriptions with different quality levels (Simple and Propose) to evaluate the impact of description quality on the performance of various model modules. Internal test set results indicated that all models performed significantly better on the optimized Propose description set compared to the Simple description set. This finding underscores the importance of generating high-quality medical descriptions using carefully designed prompts, affirming the tremendous potential of large language models to enhance the performance of medical diagnostic support systems. To thoroughly validate the adaptability of the model architecture, we systematically compared the performance of feature processing modules based on CNN, Transformer, and Mamba. By analyzing the data presented in Tables 2−4, it is evident that the Mamba module demonstrates significant advantages in cross-modal feature fusion, achieving higher accuracy compared to both the CNN and Transformer architectures. Among the various model combinations, the combination of ConvNext and RoBERTa with the Mamba feature fusion module showed the best performance, achieving the highest levels of accuracy and Kappa (accuracy: 93.7%, sensitivity: 95.8%, specificity: 94.5%, Kappa=0.89, 95% confidence interval (CI): 0.84−0.96, p<.001; Table 2).

Table 2
Comparison of mamba-based feature processing modules under variable description sets

Module	Accuracy (%)		Sensitivity (%)		Specificity (%)		Kappa (95% CI)	
	Simple	Propose	Simple	Propose	Simple	Propose	Simple	Propose
EfficientNet [28]								
textBert [29]	81.2	93.4	90.3	95.8	84.7	94.2	0.67 (0.65,0.74)*	0.85 (0.83,0.95)*
XLNet [30]	80.1	92.9	91.3	95.3	81.3	93.9	0.68 (0.64,0.73)*	0.88 (0.83,0.94)*
RoBERTa [23]	80.6	92.8	91.5	95.0	80.6	94.2	0.69 (0.64,0.73)*	0.89 (0.83,0.94)*
ResNet [31]								
textBert [29]	80.1	93.1	90.2	95.0	77.9	93.5	0.68 (0.63,0.72)*	0.89 (0.83,0.94)*
XLNet [30]	81.1	92.7	90.0	94.5	82.9	94.9	0.70 (0.66,0.74)*	0.88 (0.82,0.93)*
RoBERTa [23]	81.3	92.0	90.0	93.4	79.9	94.5	0.70 (0.66,0.75)*	0.87 (0.82,0.93)*
ConvNeXt [22]								
textBert [29]	82.4	93.4	90.6	**96.1**	84.4	93.5	0.72 (0.67,0.76)*	0.89 (0.84,0.95)*
XLNet [30]	82.3	93.0	91.4	93.2	84.0	**95.2**	0.72 (0.68,0.77)*	0.89 (0.83,0.95)*
RoBERTa [23]	81.7	**93.7**	87.4	95.8	86.1	94.5	0.71 (0.66,0.75)*	**0.89 (0.84,0.96)***

CI, confidence interval.
* p<.001.

Table 3
Comparison of transformer-based feature processing modules under variable description sets

Module	Accuracy (%)		Sensitivity (%)		Specificity (%)		Kappa (95% CI)	
	Simple	Propose	Simple	Propose	Simple	Propose	Simple	Propose
EfficientNet [28]								
textBert [29]	77.6	93.4	87.8	92.7	81.8	91.2	0.65 (0.63,0.71)*	0.84 (0.80,0.89)*
XLNet [30]	77.5	89.8	88.5	92.2	78.7	90.8	0.66 (0.62,0.70)*	0.85 (0.80,0.91)*
RoBERTa [23]	78.1	89.7	88.6	91.9	78.1	91.2	0.67 (0.62,0.70)*	0.86 (0.80,0.91)*
ResNet [31]								
textBert [29]	77.6	89.8	87.2	91.9	75.3	90.5	0.66 (0.61,0.69)*	0.86 (0.80,0.91)*
XLNet [30]	78.6	89.2	87.3	91.3	80.1	91.9	0.68 (0.64,0.71)*	0.85 (0.79,0.90)*
RoBERTa [23]	78.8	88.5	87.3	90.2	77.4	91.5	0.68 (0.64,0.72)*	0.84 (0.79,0.90)*
ConvNeXt [22]								
textBert [29]	79.9	90.1	87.8	**93.1**	81.7	92.1	0.70 (0.65,0.73)*	0.86 (0.81,0.92)*
XLNet [30]	79.8	89.7	88.5	90.2	81.2	**92.2**	0.70 (0.66,0.74)*	0.86 (0.80,0.92)*
RoBERTa [23]	79.2	**90.2**	84.8	92.7	83.4	91.5	0.69 (0.64,0.72)*	**0.86 (0.81,0.93)***

CI, confidence interval.
* p<.001.

Table 4
Comparison of CNN-based feature processing modules under variable description sets

Module	Accuracy (%)		Sensitivity (%)		Specificity (%)		Kappa (95% CI)	
	Simple	Propose	Simple	Propose	Simple	Propose	Simple	Propose
EfficientNet [28]								
textBert [29]	76.8	89.2	87.0	92.0	80.5	90.1	0.63 (0.61,0.69)*	0.80 (0.78,0.82)*
XLNet [30]	76.3	89.0	87.4	91.5	77.5	89.7	0.64 (0.60,0.68)*	0.84 (0.83,0.86)*
RoBERTa [23]	77.0	89.1	87.3	91.2	77.0	90.1	0.65 (0.60,0.68)*	0.84 (0.78,0.89)*
ResNet [31]								
textBert [29]	76.8	89.0	86.5	91.2	74.5	89.6	0.64 (0.59,0.67)*	0.84 (0.78,0.89)*
XLNet [30]	77.3	88.5	86.2	90.5	79.0	90.8	0.66 (0.62,0.69)*	0.83 (0.77,0.88)*
RoBERTa [23]	77.5	87.8	86.1	89.5	76.2	90.3	0.66 (0.62,0.70)*	0.82 (0.77,0.88)*
ConvNeXt [22]								
textBert [29]	79.2	**89.3**	86.5	**92.3**	80.5	89.6	0.68 (0.63,0.71)*	0.84 (0.79,0.90)*
XLNet [30]	78.9	88.9	87.2	89.5	80.0	**91.1**	0.68 (0.64,0.72)*	0.84 (0.78,0.90)*
RoBERTa [23]	78.5	89.1	83.8	91.9	82.1	90.3	0.67 (0.62,0.70)*	**0.84 (0.79,0.91)***

CI, confidence interval.
* p<.001.

Comparative performance with other methods

The GPT4LFS classification results for the internal test set were compared to those of the current state-of-the-art (SOTA) methods (Table 5). The results indicated that GPT4LFS significantly outperformed existing technologies across all performance metrics. Specifically, GPT4LFS achieved accuracy of 93.7%, sensitivity of 95.8%, and specificity of 94.5%, which far exceeded the best-performing SOTA model MedMamba, with accuracy, sensitivity, and specificity rates of 82.4%, 79.5%, and 80.0%, respectively. This clearly demonstrated the superior capabilities of GPT4LFS in classification tasks. Additionally, the model exhibited extremely high consistency and reliability (Kappa=0.89, 95% CI: 0.84−0.96, p<.001). These results

Table 5
Comparative performance with other methods

Model name	Accuracy (%)	Sensitivity (%)	Specificity (%)	Kappa (95% CI)
Spine-AI [2]	52.4	53.5	51.1	0.47 (0.45,0.51)*
yolov5 [32]	62.4	61.2	60.1	0.56 (0.53,0.61)*
yolov8 [25]	71.3	71.1	69.9	0.61 (0.59,0.65)*
Swin-transformer [26]	72.4	73.9	71.1	0.64 (0.62,0.78)*
MedMamba [27]	82.4	79.5	80.0	0.74 (0.73,0.78)*
GPT4LFS (ours)	**93.7**	**95.8**	**94.5**	**0.89 (0.84,0.96)***

CI, confidence interval.
* p<.001.

validated the exceptional performance and broad application potential of GPT4LFS in medical image analysis tasks.

Model evaluation using internal and external test sets

One typical foraminal image was selected from each layer as a benchmark to ensure data accuracy and representativeness. This resulted in an internal test set composed of 820 images and an external test set consisting of 930 images, with validation conducted for both data groups (Table 6). In both the internal and external test sets, the number where the model (Mod) prediction and the doctor (Doc) diagnosis were in agreement was recorded, along with the consistency of these results (shown in parentheses). The model's accuracy in identifying the L_5-S_1 level was the lowest in both test sets, which may be due to the variability in pelvic tilt, resulting in the model's reduced ability to recognize images at this level compared to other levels. Overall, the model's performance in recognizing the severe level in the external test set was inferior to its performance in the internal test set, which may be attributed to the image data differences.

Furthermore, consistency tests were conducted on the 2 test sets separately. The model achieved an accuracy of 93.7%, sensitivity of 95.8%, and specificity of 94.5% for the internal test set. The model's LFS classification showed excellent consistency with the orthopedic doctors' standards (Kappa=0.89, 95% CI: 0.84−0.96, p<.001). The model achieved an accuracy of 92.2%, sensitivity of 92.2%, and specificity of 97.4% for the external test set. The model's LFS classification also showed excellent consistency with the orthopedic doctors' standards (Kappa=0.88, 95% CI: 0.84−0.89, p<.001). These results indicated that the model's overall performance on both the internal and external test sets was excellent, demonstrating its good generalization ability when faced with new data. Even with unfamiliar datasets, the model can still effectively assist doctors in making accurate diagnoses.

Discussion

Lumbar MRI scanning is an important tool for assessing LFS, as it can accurately describe stenosis severity [21]. However, in the past, the diagnostic process has relied on the subjective interpretation of radiologists, which is not only time-consuming but also prone to significant interpretative discrepancies. To address this issue, the present study developed an automated assistance diagnostic model based on MLLM (GPT4LFS) for automatic detection of the degree of LFS in MR images.

GPT4LFS demonstrated excellent consistency in the 4-class classification task (normal, mild, moderate, or severe) in the internal test set, achieving excellent consistency (Kappa=0.89, 95% CI: 0.84−0.96, p<.001) based on only 6,299 MR images. The external test set was evaluated using datasets from 2 different regions to validate the model's generalization capability. Testing results also indicated that the model achieved excellent consistency in classifying the degree of LFS relative to the standards set by the orthopedic doctors (Kappa=0.88, 95% CI: 0.84−0.89, p<.001). This outcome was significantly better than the performances described by Hallinan et al [2] (Kappa=0.75) and Lim et al [9] (Kappa=0.71), which used 18,564 MR images, showcasing the model's exceptional performance and potential. Moreover, the performance of GPT4LFS on both the internal and external test sets was nearly consistent, indicating that the model can effectively generalize without overfitting specific datasets. This is particularly important for clinical applications.

Furthermore, the present study model demonstrated superior performance compared to that of the current SOTA image classification models. The consistency of YOLOv8 [25] was relatively low (Kappa=0.61, 95% CI: 0.59−0.65, p<.001), primarily because its design focus is on real-time object detection and is not optimized for the subtle features present in medical imaging. While Swin-Transformer [26] exhibits excellent performance on general datasets, it still has limitations in extracting complex features from medical images (Kappa=0.64, 95% CI: 0.62−0.78, p<.001). MedMamba [27] is specifically designed for medical images and showed better consistency (Kappa=0.74, 95% CI: 0.73−0.78, p<.001) than the other models, although it still falls short of the requirements for practical clinical applications. In contrast, the performance of GPT4LFS (Kappa=0.89, 95% CI: 0.84−0.96, p<.001) was undoubtedly more advantageous, reflecting the potential of MLLM in the field of medical imaging analysis. In comparison, the present study results did not only significantly surpass those of traditional deep learning models but also provided a robust reference framework for future research.

The proposed study described several key innovations: (1) The incorporation of medical description text generated by MLLM, which was encoded and feature-extracted using RoBERTa, providing the model with rich contextual information. These medical descriptions containing detailed image accounts and relevant patient information, enhanced the model's feature representation and significantly improved its contextual understanding capabilities; (2) The use of ConvNeXt as the image processing module, focusing on the extraction of high-dimensional imaging features. This was because medical images often contain a wealth of complex detailed information, and the model can more comprehensively represent the pathological patterns in the images by capturing these high-dimensional features, significantly improving its performance on classification and understanding tasks; (3) The adoption of the Mamba architecture to effectively integrate image and text features, improving the model's understanding of complex relationships and enhancing the richness of the feature representation, which significantly improved the accuracy and robustness of the classification. This multimodal approach optimized the contextual understanding capabilities and

Table 6
Model's evaluation of LFS degree at different levels in the internal and external test sets

Classification	L_{1-2}		L_{2-3}		L_{3-4}		L_{4-5}		L_5-S_1	
	Doc	Mod	Doc	Mod	Doc	Mod	Doc	Mod	Doc	Mod
Internal										
Normal	121	119 (98.3%)	75	72 (96.0%)	37	34 (91.9%)	13	11 (84.6%)	48	39 (81.3%)
Mild	41	36 (87.8%)	79	75 (94.9%)	104	100 (96.2%)	91	86 (94.5%)	73	69 (94.5%)
Moderate	2	2 (100%)	9	8 (88.9%)	19	16 (84.2%)	50	47 (94.0%)	26	23 (88.5%)
Severe	0	0 (100%)	1	1 (100%)	4	4 (100%)	10	9 (90.0%)	17	16 (94.1%)
Total	164	157 (95.7%)	164	156 (95.1%)	164	154 (93.9%)	164	153 (93.3%)	164	147 (89.6%)
External										
Normal	107	105 (98.1%)	39	35 (89.7%)	7	7 (100%)	1	1 (100%)	41	37 (90.2%)
Mild	69	64 (92.8%)	125	122 (97.6%)	136	133 (97.8%)	95	93 (97.9%)	58	52 (89.7%)
Moderate	7	7 (100%)	20	15 (75.0%)	35	29 (82.9%)	60	56 (93.4%)	56	47 (83.9%)
Severe	3	3 (100%)	2	0 (0%)	8	3 (37.5%)	30	21 (70.0%)	31	26 (83.9%)
Total	186	179 (96.2%)	186	172 (92.5%)	186	172 (92.5%)	186	171 (91.9%)	186	162 (87.1%)

Mod, model prediction; Doc, doctor diagnosis.

enhanced the model's ability to process complex medical data.

While this study made significant progress in the analysis of lumbar MR images, there is still room for improvement. First, the study focused on sagittal MR lumbar foramen images, and integration of additional imaging modalities in the future, such as axial and coronal MR images, could capture more comprehensive pathological information. Second, the high cost of using the GPT-4o large language model limited the ability to conduct larger-scale tests, which served as the motivation for model optimization. To comprehensively validate the model's clinical applicability, prospective studies are needed to more accurately assess its performance in actual clinical environments.

Overall, the MLLM-based diagnostic assistance method proposed in the study demonstrated excellent performance in improving diagnostic efficiency and accuracy, showing promising clinical application prospects. Additional research and optimization, particularly prospective studies and the integration of more imaging modalities, may further enhance the clinical value and applicability of the model. These efforts will help the model to play a greater role in real-world medical practice, providing patients with more accurate and comprehensive diagnostic support.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Elzat Elham-Yilizati Yilihamu: Writing − review & editing, Writing − original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fan-Shuo Zeng:** Writing − review & editing, Writing − original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Jun Shang:** Writing − review & editing, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation. **Jin-Tao Yang:** Writing − review & editing, Writing − original draft, Visualization, Validation, Software, Methodology, Conceptualization. **Hai Zhong:** Writing − review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shi-Qing Feng:** Writing − review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Data sharing statement

Data generated or analyzed during the study are available from the corresponding author by request.

Ethics approval and consent to participate

This study was approved by the Ethics Committee of the Tianjin Medical University General Hospital (IRB2022-YX-213-01), the Second Hospital of Shandong University (KYLL-2023-324) and Xuzhou Renci Hospital (XZRCKY-KT-202306001).

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Supplementary materials

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